

Consumer-Aligned Learning for Quality-of-Transmission: The False-Clear Rate at the FEC Cliff

Andrew H. Bond

Department of Computer Engineering, San José State University, San José, CA 95192, USA
andrew.bond@sjsu.edu

Rendering preview (standard article class). The official OFC 2027 submission is typeset with the Optica opticameet3 meeting class; this document reproduces the identical content and figures for review and sharing. All numbers are sealed and graded (XPROTO-QOT and XPROTO-QOT-ML, 2026-08-25) or deterministic.

Abstract: An ML-QoT estimator trained on average GSNR error wins on average and fails at the FEC threshold. On GNPY over CORONET-CONUS, training for the decoder’s threshold cuts false-clears five times over at equal capacity.

1 Introduction

Machine-learning quality-of-transmission (ML-QoT) estimation is a leading route to reducing the system margin that optical operators carry against uncertain reach performance [2, 3]. That margin is deliberately stranded capacity, so every fraction of a dB matters. ML-QoT models are almost always trained to minimise a prediction error, mean-squared or mean-absolute GSNR error. We argue, and then measure, that this objective serves the wrong master. The consumer of a QoT estimate is the format-selection decision, and behind it the FEC decoder, and the decoder reads a threshold. A lightpath either meets its format’s required GSNR or it does not. An estimator can be excellent on average and still, near a format boundary, over-predict by just enough to select a rate the link cannot support. The result is an in-service outage even though the mean error was small. We call this a false-clear.

We measure the false-clear rate on the GN-model physical layer of GNPY [1] over the CORONET-CONUS reference network and show three things. An average-error ML-QoT objective false-clears at the FEC cliff while a threshold-aware objective does not, at equal capacity. Which lightpaths are at risk is set by spectral position and reach, so no single blanket margin fits all of them. And the risk grows with the fill added after provisioning, which gives the certificate a refresh horizon.

2 Method

Substrate. Lightpaths are transparent shortest-path routes of the CORONET-CONUS reference topology (75 nodes; up to 2500 km, 2–31 spans of 80 km). Per-channel GSNR comes from the GN model in GNPY [1] on standard single-mode fibre (0.2 dB/km, $D = 17$ ps/nm/km) with 6 dB-NF EDFAs, +2 dBm per channel, 32 GBd on a 50 GHz grid. The band runs from a lightly filled provisioning state of 6 channels to a full C-band of 76.

Certificate, consumer, witness. A certificate selects the highest format whose required GSNR the estimate clears. The consumer is that decision. The witness is the true deployed

GSNR, in practice the receiver’s pre-FEC BER. For the ML study the features are the provisioning-time GSNR plus spectral position, reach and fill, the label carries an irreducible term for unmeasured network state, and two gradient-boosted objectives compete: average-error (squared loss) and consumer-aware (a conservative lower-quantile loss that protects the decision rather than the mean). Bars and manipulation checks are pre-registered. The sealed record is content-hashed and graded on seeds disjoint from those used during design [5].

3 Results

3.1 ML-QoT trained on the wrong metric

Under irreducible prediction uncertainty the average-error estimator wins the metric it was trained on and loses the one that matters. Its MAE is 0.65 dB against 0.95 dB for the consumer-aware objective, yet its false-clear rate is about 0.041 against 0.008, five times higher, at comparable delivered spectral efficiency near 3.9 bits/symbol. Figure 1(a) shows the two objectives trading places between the two metrics. Figure 1(b) shows where the failures live: the average-error false-clears concentrate where the GSNR headroom above the nearest format threshold is small. That is the cliff, and it is exactly where a symmetric error metric is blind. The fix is to train ML-QoT on the consumer’s threshold decision rather than on volume-averaged fidelity.

3.2 The margin is consumer-relative

The same threshold-blindness afflicts the margin itself as the band fills. A lightpath’s GSNR drops as neighbours arrive, and whether it crosses a format threshold depends on where it sits: false-clears concentrate on services whose spectral position and reach leave their GSNR near a boundary, and are absent elsewhere (Fig. 2). So a single blanket margin cannot be right for every lightpath. We price this against today’s practice of one uniform system margin sized to keep the whole fleet safe. The deployed fill is real. Each lightpath’s channel

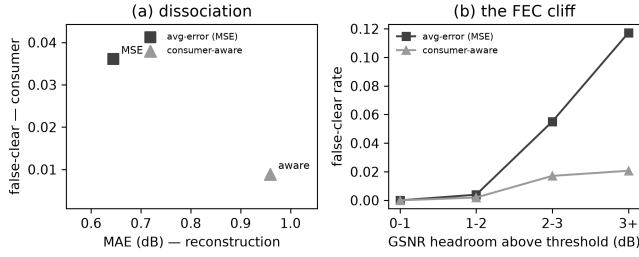


Figure 1: Two objectives, two metrics (GNPy, gradient-boosted regressors). (a) The average-error objective has lower MAE and higher false-clear than the consumer-aware objective. (b) Its false-clears sit at the FEC cliff, where headroom is small. Marker shape distinguishes objectives.

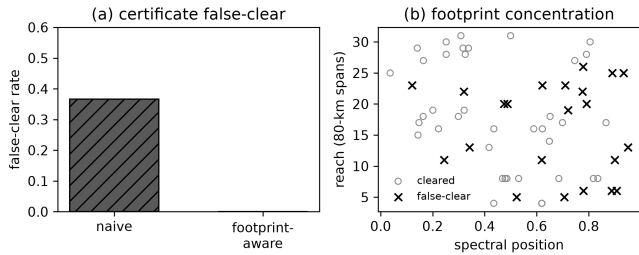


Figure 2: Where the margin fails (GNPy over CORONET-CONUS). Per-service false-clear over spectral position and reach concentrates on threshold-adjacent services. A footprint-aware certificate removes it. Marker shape distinguishes classes.

load is drawn from the per-link fill distribution of the SNDlib janos-us-ca demand matrix [4] (61 links; median 10 channels, 90th percentile 35, of 76). Real traffic is heavy-tailed. A uniform margin sized for the few near-full links over-provisions the lightly filled majority. We give both policies the same QoT-estimator uncertainty of 0.5 dB RMSE, so the footprint-aware certificate is not assumed perfect. Sizing each margin to the lightpath’s own fill then recovers about 8% of spectral efficiency, near 0.33 b/sym or 21 Gb/s per channel, at matched safety (false-clear at most 0.01) on a rate-flexible transceiver model. The gain is modest and free. It is capacity the uniform margin strands on links that never fill.

3.3 The refresh floor

The risk is driven by the fill added between provisioning and use, so a certificate has a refresh horizon. Sweeping the deployed fill from 6 to 76 channels, the naive false-clear rises roughly linearly with the number of added channels, $FC \approx 0.008\Delta N$ with $R^2 = 0.95$. At a 0.05 target the floor is about 8 added channels. The rule is concrete. Re-certify a service before the band around it grows by about 8 channels, rather than on a fixed calendar.

4 Related work

QoT estimation and physical-layer-aware routing and spectrum assignment are mature [1, 3]. ML-QoT is active, and its estimators are trained mostly on prediction error [2]. We do not offer a better estimator or allocator. We offer the measurement, the false-clear rate read against the consumer, and the finding that both the ML-QoT training objective and the margin allocation should target the decoder’s threshold.

5 Discussion

Train QoT models, and size QoT margin, for the decision the FEC decoder makes rather than for average fidelity. The evidence stands on the GN model. A coherent-receiver pre-FEC-BER witness on a fibre testbed, and the EGN or split-step model, are the natural next steps. The method is not specific to optics. The same consumer-relative false-clear appears in radio link adaptation and in database replication [5]. The results here stand on the optical QoT certificate alone.

References

- [1] A. Ferrari *et al.*, “GNPy: an open source application for physical layer aware open optical networks,” J. Opt. Commun. Netw. **12**, C31–C40 (2020).
- [2] C. Rottondi *et al.*, “Machine-learning method for quality of transmission prediction of unestablished lightpaths,” J. Opt. Commun. Netw. **10**, A286–A297 (2018).
- [3] Y. Pointurier, “Design of low-margin optical networks,” J. Opt. Commun. Netw. **9**, A9–A17 (2017).
- [4] S. Orlowski, R. Wessälly, M. Pióro, and A. Tomaszewski, “SNDlib 1.0—Survivable Network Design Library,” Networks **55**, 276–286 (2010).
- [5] A. H. Bond, “Observation Theory: consumer-relative witnessed certificates,” sealed pre-registrations and code, <https://github.com/ahb-sjsu/observation-theory-campaigns> (2026).